Victor Jean Viaunel

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RESEARCH INTERESTS

Scan-to-BIM with mobile LiDAR · as-built BIM for existing buildings and energy retrofit · IFC and openBIM workflows · language models for BIM generation · computer vision for construction data

EDUCATION

2024 – 2026 — **M.S. Civil Engineering** · National Taiwan University
H-Team lab (BIM / IFC / Digital Construction), advised by Prof. Shang-Hsien (Patrick) Hsieh. Research area: scan-to-BIM with mobile LiDAR.

2019 – 2023 — **B.S. Civil Engineering** · National Yang Ming Chiao Tung University

PUBLICATIONS

2026 · Proceedings — **Victor, J. V.**, Kumar, P., & Hsieh, S.-H. (2026). From 2D drawings to BIM models: An LLM-assisted approach. In *Proceedings of the 43rd International Symposium on Automation and Robotics in Construction (ISARC)* (pp. 2410–2414). Singapore: IAARC. doi:10.22260/ISARC2026/0307

2026 · Conference — **Victor, J. V.**, Chang, Y.-T., & Hsieh, S.-H. (2026). A review on scan-to-BIM for building energy retrofit. *International Conference on Computing in Civil and Building Engineering (ICCCBE 2026)*.

PRESENTATIONS

25 Mar 2026 — "A Review on Scan-to-BIM for Building Energy Retrofit." ICCCBE 2026, Session 8B: Infrastructure Systems. Presenter.

RESEARCH AND TEACHING

Current — **Assistant Researcher** · H-Team lab, National Taiwan University
Research on scan-to-BIM and LLM-assisted BIM modelling, from point-cloud processing to IFC export.

Current — **Teaching Assistant** · BIM course, Prof. Shang-Hsien Hsieh, NTU
Support students with practical BIM topics and help them develop their own implementation ideas.

Jun – Aug 2022 — **Research Intern, Computer-Aided Engineering** · National Taiwan University
Computer vision project in OpenCV and PyTorch, plus sequence models with LSTMs and GRUs.

PROFESSIONAL EXPERIENCE

2024 – Present — **Lead Full-Stack Developer** · Legend Castle, Taiwan
Main developer of MaiSchool, a live online tutoring platform for families, from design to implementation (Next.js, Convex, Clerk, Zod).

Current

Founder · GUANIN (Guanin Co., Ltd.), Taiwan
Sourcing marketplace connecting suppliers in Asia with buyers in the Caribbean and Latin America; built and run the platform.

Feb – Jun 2023

Structural Engineering Intern · Envision
Assistant structural engineer; structural modelling and analysis in ETABS and SAP2000.

COMPETITIONS

2026

International AI Challenge on Marine Debris Image Recognition · Taiwan
15th of 69 teams in the preliminary round (two-person team), macro mAP@0.5 of 0.7917 across 34 debris classes.

2026

AI CUP: Solar Panel Defect Detection · Ministry of Education, Taiwan
Public score improved from 0.301 to 0.8337; about 29th of 121 teams.

SERVICE AND LEADERSHIP

Current · 2nd term

Academic Committee Lead · Haitian student community in Taiwan
Guide bachelor's to graduate applicants through Taiwanese university admissions: study plans, autobiographies, documentation, and application workshops that I coordinate.

SKILLS

BIM and engineering	Revit, IFC, AutoCAD, ETABS, SAP2000, SAFE
Reality capture	Mobile LiDAR scanning, point-cloud processing, photogrammetry
Computing	Python, PyTorch, OpenCV, YOLO, LLM agents, Next.js, Convex, MongoDB, Docker